

MAT 426, Class 14 Homework: Householder Matrices
(Due Thursday, April 2, 2026)

1. Let H be a Householder matrix with nonzero vector $\mathbf{u} \in \mathbb{R}^n$. Prove each of the following.

(a) $\|H\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$

(b) $H^2 = I$

(c) $H\mathbf{u} = -\mathbf{u}$

(d) $H\mathbf{v} = \mathbf{v}$ if $\mathbf{v}^T\mathbf{u} = 0$

2. Determine the eigenvalues of a Householder matrix H and their algebraic multiplicities. Use this to determine the determinant of H .

3. For the given nonzero vector $\mathbf{u}$, find the Householder matrix H .

(a) $\mathbf{u} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(b) $\mathbf{u} = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$

(c) $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

4. For each of the vectors $\mathbf{x}$ below, create a Householder matrix H so that $H\mathbf{x}$ is a multiple of $\mathbf{e}_1$.

(a) $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$

(b) $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

(c) $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

5. Consider a nonzero vector $\mathbf{u} \in \mathbb{R}^2$. We want to construct a matrix H such that $H\mathbf{u} = -\mathbf{u}$ and reflects all other vectors through the line perpendicular to $\mathbf{u}$.

(a) Suppose $\mathbf{u}$ is in the second quadrant (between 90° and 135°). On the same set of axes, sketch $\mathbf{u}$, the line through the origin perpendicular to $\mathbf{u}$, a vector $\mathbf{x}$ in the first quadrant (between 45° and 90°), and its reflection $H\mathbf{x}$.

(b) Express $H\mathbf{x}$ in terms of $\mathbf{u}$ and $\mathbf{u}_\parallel$ (the part of $\mathbf{x}$ that is parallel to $\mathbf{u}$).

(c) Rewrite the expression of $H\mathbf{x}$ from (b) in terms of dot products.

(d) Use the fact that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$ to rewrite the expression from (c). Identify this expression.